Important chemical reactions



Oxidation

$4\ Fe + 3\ O_2 \rightarrow 2\ Fe_2O_3$

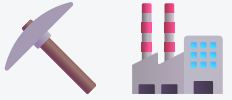
A substance reacts with the oxygen in the air. Iron rusts, especially with moisture.



Combustion

fuel + O_2 → CO_2 + H_2O + energy

A fast oxidation that releases a lot of heat and light (butane, petrol, wood).



Obtaining metals

minerals → metal

Iron from iron oxides in a blast furnace; aluminium from bauxite; copper from chalcopyrite.



Photosynthesis & respiration

CO_2 + H_2O ⇄ glucose + O_2

Plants store the Sun's energy in glucose; living things release it by breathing.



Fermentation

sugars → alcohol or acid + CO_2...

Made by yeasts and bacteria: bread, yoghurt, cheese, wine.



Neutralisation

acid + base → salt + water

HCl + NaOH → NaCl + H_2O. Antacids neutralise stomach acid.